When Trends Become Policy: Online Attention, Consumer Goods, and Regulation in China

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***Abstract*—This paper examines how high-visibility online controversies in China are associated with real-world market shifts and policy responses. Rather than presenting a direct China–U.S. empirical comparison, the paper uses China as the main empirical setting and draws on U.S.-based scholarship as a comparative baseline from prior literature. The central question is whether platform-level attention controls in China shape the threshold and pathway through which online discussion becomes a visible firestorm, and whether episodes that cross that threshold are followed by observable spillovers in markets or governance. To explore this question, the paper combines literature review with four illustrative China-focused cases: the 2024 Nongfu Spring boycott crisis, the 2023 Japan nuclear wastewater salt panic, the 2022 Urumqi residential fire and Zero COVID policy reversal, and the late-2022 Lianhua Qingwen medicine-demand surge during reopening. The empirical analysis aligns attention proxies, stock-market indicators, and event timelines to identify patterned associations between peak attention and subsequent offline outcomes. Across the four cases, the paper compares reputational boycott pressure, panic buying, protest-linked policy reversal, and reopening-driven pharmaceutical repricing. The goal is not to make strong causal claims, but to clarify a cautious framework linking platform governance, public attention, and spillover under conditions of partial information.**

I. Introduction

A. Background

Social media platforms do not simply host public attention; they help organize it. Ranking systems, recommendation algorithms, search interfaces, moderation practices, and trending lists all affect whether discussion remains dispersed or becomes legible as a broad public controversy. This question is especially important in China, where platform governance operates within a denser regulatory environment and where recommendation systems are themselves explicit objects of information governance [1]–[3]. In that setting, the relevant issue is not whether criticism or rumor exists at all, but how platform-level attention controls shape the conditions under which online discussion becomes widely visible.

Prior scholarship suggests that Chinese information control is selective rather than total. King, Pan, and Roberts argue that censorship is aimed less at criticism as such than at forms of expression that may enable collective action or coordinated escalation [1]. Gallagher and Miller similarly show that information control often depends on who is speaking, not only on what is being said [2]. Comparative work by Kim et al. further suggests that online firestorms in China face a higher threshold of activation than comparable episodes in the United States [4]. Taken together, these studies imply that platform governance may change the visibility, escalation, and spillover conditions of contentious attention rather than fully determining public opinion.

B. Motivation

This topic matters for three reasons. First, online attention can spill beyond the platform itself. Prior work links social media sentiment to financial outcomes in both U.S. and Chinese settings, and other research shows that online scandal exposure in China is associated with quicker official response and more severe punishment [5]–[7]. Related work on TikTok also suggests that platform attention can shape behavioral intentions through perceived opinion leadership and hedonic engagement, even when the outcome is consumer response rather than policy change [8]. These studies do not establish a single causal mechanism, but they do indicate that social-media visibility can become economically and politically consequential.

Second, China provides a useful setting for studying how platform design and public attention interact under stronger institutional oversight. The point is not to assume that the state or platforms fully control public reaction. Rather, the Chinese case makes it easier to examine how visibility is filtered, delayed, or amplified under more formalized governance conditions. Third, a China-centered design helps separate two questions that are often conflated: whether criticism exists, and under what conditions that criticism becomes large, visible, and sustained enough to coincide with market or governance responses.

C. Research Question

This paper asks: *How do platform-level attention controls in China shape the threshold and pathway through which online discussion becomes a high-visibility firestorm, and once that threshold is crossed, how consistently are market or policy spillovers observed?* A secondary comparative question uses the U.S. literature as a baseline rather than as a matched empirical case: are the China cases consistent with prior arguments that more centrally governed platform environments produce fewer visible firestorms, but potentially more consequential ones when they do emerge [4]?

This framing organizes the analysis around three linked ideas. The first is *threshold*: when does routine online discussion become a public-attention event large enough to matter? The second is *pathway*: through what mechanisms does a topic move from scattered posts to concentrated visibility? The third is *spillover*: once attention becomes sufficiently visible, what kinds of market or governance consequences follow, and on what timeline? Framed this way, the paper treats China as the main empirical setting and uses the United States primarily as a comparative reference point from prior scholarship. Across all four cases, the analysis asks the same five questions: what triggered visibility, how threshold crossing is identified, what spillover followed, how quickly it followed, and what type of mechanism the sequence appears to represent.

D. Contributions

This paper makes four more modest contributions. First, it proposes a conceptual vocabulary that separates threshold, pathway, and spillover instead of treating "virality" or "public opinion" as a single outcome. Second, it brings together literatures on online firestorms, Chinese information control, and social-media spillovers that are often discussed separately. Third, it applies that framework to four China-based illustrative cases that capture analytically distinct pathways: a reputational boycott and slow-burn market damage, a panic-buying commodity spike, a protest-pressure policy reversal with reopening repricing, and a reopening-driven medicine-demand shock with product-linked pharmaceutical reaction. Fourth, it clarifies the limits of inference in the current design: because the evidence is case based, proxy dependent, and not paired with a direct U.S. case, the paper is designed to identify patterned associations and plausible mechanisms rather than to make strong causal claims.

II. Related Work

A. Online Firestorms and Public Attention

Research on online firestorms examines how concentrated waves of negative attention emerge, intensify, and spread across social media. For this paper, the most directly relevant study is Kim et al., who compare trending topics and associated posts on China Weibo and U.S. Twitter over a six-month period [4]. They find that the threshold for classifying an episode as a firestorm appears higher in China than in the United States, and they also show that the targets and expressive styles of firestorms differ across the two settings. That finding is important here because it suggests that platform environment affects not only how much criticism is expressed, but also which disputes become legible as large-scale public events.

At the same time, much of the firestorm literature remains centered on the online episode itself. It is strong on questions of escalation, framing, and audience participation, but it is less likely to trace how attention shocks relate to market behavior, consumer response, or official intervention in a single framework. The present paper builds on this literature by keeping firestorm formation in view while following its possible consequences beyond the platform.

B. Content Moderation, Censorship, and Platform Governance

Chinese platform governance research helps explain why threshold dynamics may differ across settings. King, Pan, and Roberts provide the foundational argument that censorship in China is not primarily organized around suppressing criticism of the state; instead, it is more consistently directed at speech that may facilitate collective action or coordinated expression [1]. Gallagher and Miller refine this perspective by arguing that China's information control strategy often depends on the identity and political position of the speaker as much as on the content of the message [2]. These accounts move the discussion away from a simple state-versus-criticism model and toward a more specific question: how do moderation and visibility rules alter the conditions under which online attention escalates?

This governance logic is also visible in formal regulation. The 2022 *Internet Information Service Algorithmic Recommendation Management Provisions* explicitly cover recommendation systems, ranking and selection systems, and search-related services, indicating that visibility itself is treated as a governable object [3]. For the present paper, this matters because it supports a platform-governance perspective in which attention is structured not only by users and networks, but also by institutionalized rules that affect discoverability, amplification, and persistence.

C. Social Media Spillovers to Markets and Policy

Another literature asks whether online attention predicts or accompanies outcomes beyond the platform. In financial contexts, Adams et al. show that overnight Twitter financial sentiment helps predict next-day stock returns in the United States, while Xu et al. find time-varying relationships between Weibo sentiment and stock returns in China [5], [6]. These studies caution against reading sentiment as a simple one-way driver, since market movements can also feed back into online discussion. Even so, they establish that digital attention and market behavior often move together in substantively meaningful ways.

Research on governance spillovers points in a similar direction. Chen and Li find that higher levels of online scandal exposure in China are associated with quicker government response and more severe punishment of officials [7]. Although this literature focuses on political scandals rather than consumer-goods controversies, it is directly relevant because it shows that once online visibility reaches a sufficient scale, it can coincide with formal institutional response. Finally, TikTok-related marketing research suggests that platform exposure can affect behavioral intentions through attention and perceived influence [8]. This does not make TikTok evidence directly comparable to Chinese crisis cases, but it helps support the broader claim that online visibility can spill into off-platform behavior.

D. Gap in Existing Literature

Existing scholarship therefore establishes several pieces of the puzzle, but usually in isolation. One set of studies explains how firestorms form. Another explains how Chinese information control works. A third shows that social-media signals can be associated with market or governance outcomes. What remains less developed is an integrated framework that connects these three dimensions without overstating causal certainty.

This paper addresses that gap by asking not only whether online trends matter, but when they become visible enough to matter, how platform governance shapes the route to that visibility, and whether threshold-crossing events are followed by observable spillovers in markets or policy. In that sense, the paper treats firestorms, information control, and offline consequences as sequentially related parts of the same process rather than as separate topics.

III. Conceptual Framework

A. Threshold

The first concept is *threshold*. In this paper, threshold refers to the point at which ordinary online discussion becomes sufficiently concentrated and visible to qualify as a high-attention public event. Threshold should not be defined only by absolute volume, because platforms differ in baseline activity, moderation practices, and visibility rules. Instead, threshold is best understood as an abnormal concentration of attention relative to platform and case context.

To avoid a purely post hoc definition, the current draft uses a simple operational rule. A case is treated as having crossed the visibility threshold when at least two of the following three indicators appear within the same one- to two-week window: *(1)* a major search-interest surge relative to the recent baseline, defined here as either a threefold increase over the prior four-week average or an observation above 60 on the normalized Google Trends scale; *(2)* evidence of breakout attention, such as multiple "Breakout" related queries or contemporaneous reporting that the topic had become viral on Weibo; and *(3)* an accompanying sign of abnormal offline attention, such as a sharp increase in investor-search terms, trading volume, or immediate official response. This rule is necessarily heuristic, but it makes the timing logic explicit and keeps threshold from being assigned only after the fact.

The same two-of-three rule is applied consistently across all four cases. It is still a heuristic, however, and is intended to reduce post hoc labeling rather than eliminate judgment entirely. The available evidence comes mainly from attention proxies and event timing rather than from complete native-platform logs. Threshold is therefore treated as an observable shift in public visibility, not as a perfectly measured latent state. This definition is consistent with prior comparative firestorm work while remaining sensitive to the specific constraints of China-focused data [4].

B. Pathway

The second concept is *pathway*. Pathway refers to the process through which an issue travels from scattered online discussion to concentrated visibility and then to possible offline consequences. In simplified form, the pathway runs from an initial trigger, to user discussion, to amplification or suppression dynamics, to a threshold-crossing visibility event, and finally to market or governance response.

In China, this pathway is likely mediated by platform-level attention controls, including moderation practices, recommendation logic, search filtering, and differentiated treatment of speakers or account types [1]–[3]. For that reason, pathway is not reducible to organic virality alone. Two issues with similar substantive content may follow different trajectories depending on whether influential accounts participate, whether the discussion is interpreted as socially risky, and whether the topic remains visible long enough to attract broader response. Because the present paper uses case evidence rather than internal platform logs, these mechanisms are inferred cautiously rather than observed directly.

C. Spillover

The third concept is *spillover*. Spillover refers to observable consequences outside the platform that are temporally associated with periods of peak online attention. This paper focuses on two forms.

The first is *market spillover*, including stock-price movement, abnormal trading volume, boycott behavior, consumer hoarding, and other measurable shifts in market-relevant activity. The second is *policy or governance spillover*, including official statements, regulatory announcements, enforcement actions, and other institutional responses. These outcomes should not be collapsed into a single category, because they may differ in both timing and mechanism. Market spillovers may emerge quickly through expectation, rumor, or consumer reaction, whereas policy spillovers may reflect slower institutional processing even when they occur within the same attention window.

D. Framework Logic

The framework proposed here is sequential and conditional. Platform governance affects the probability that an issue crosses a visibility threshold, the route by which it does so, and the kinds of spillovers that become more or less likely once visibility is achieved. The claim is therefore not that governance fully determines attention or that every online controversy in China produces strong offline effects. The more defensible claim is narrower: platform governance changes visibility, escalation, and spillover conditions.

This logic also helps explain why China may show both fewer visible firestorms and, at times, more consequential ones once they do emerge. If high-visibility attention events are more selectively produced, then crossing the threshold may itself carry informational value for firms, investors, regulators, and consumers. In that sense, visibility in a more governed

environment may be rarer, but it may also be more salient when it occurs.

E. Hypotheses

- H1: Consistent with prior literature, firestorm thresholds are higher in China than in less centrally governed U.S. platform environments.
- H2: In the China cases examined here, episodes that cross a clear visibility threshold are more likely to coincide with observable market or governance spillovers than periods that remain below that threshold.
- H3: When spillovers occur, market responses tend to appear faster than formal policy responses, although both are most likely around peak public-attention windows.

These hypotheses are assessed in different but complementary ways. H1 is evaluated indirectly, by comparing the China cases to the comparative baseline established in prior U.S.–China firestorm scholarship rather than by building a new matched U.S. dataset. H2 is evaluated by locating threshold-crossing windows in each case and then asking whether observable market or policy spillovers cluster around those windows rather than around routine background periods. H3 is evaluated by comparing the timing of market and policy responses across the four cases, including whether market repricing appears during the attention burst, after it, or only once formal policy action occurs.

IV. Data and Case Selection

A. Case Selection Logic

The current empirical design is intentionally narrow and illustrative rather than comprehensive. The paper centers on four China-focused cases with clear event triggers, measurable public-attention spikes, and observable market or policy consequences: the 2024 Nongfu Spring public-opinion crisis, the 2023 Japan nuclear wastewater salt panic, the 2022 Urumqi residential fire and subsequent Zero COVID policy reversal, and the late-2022 Lianhua Qingwen medicine-demand episode during reopening. Together, these cases widen the range of mechanisms under comparison without pretending to solve the generalization problem.

The first case is a slow-burn reputational controversy in which online hostility toward a consumer brand coincides with prolonged market deterioration. The second is a short, high-intensity panic episode in which online fear, panic buying, and immediate state response unfold almost simultaneously. The third adds a politically charged pathway in which online attention and protest pressure are associated with rapid policy reversal and then with recovery in reopening-sensitive stocks. The fourth case was selected because it combines a clear policy trigger, a measurable search-attention shock, and an observable stock-market response, while operating through a distinct mechanism: reopening-driven health-information demand and product-linked pharmaceutical repricing rather than boycott pressure, commodity hoarding, or protest-to-policy reversal.

Framed this way, the cases are not a balanced sample of all Chinese online controversies; they are *illustrative* or *most-likely* cases chosen to compare distinct spillover pathways under a common threshold framework. This case-selection strategy should not be mistaken for a population-level test. Even with four cases, the design remains theory-building and process-tracing oriented. Its value lies in clarifying how threshold, pathway, and spillover may work in practice while keeping the limits of generalization explicit.

B. Data Sources

The empirical dataset combines three main components: stock-market data, public-attention proxies, and event timelines. Stock data were collected from Investing.com historical price pages. Public-attention data were collected from Google Trends using paired keyword series for each case. Event timelines were compiled from financial news reports, government announcements, corporate disclosures, and contemporaneous reporting on Weibo virality, panic behavior, protest escalation, or reopening-related medicine demand.

Google Trends is used here as an externalized search-attention proxy around already salient events, not as a direct measure of native Weibo or Douyin visibility. That distinction matters, especially in China, where platform-level visibility is shaped by recommendation systems, moderation, and search filtering that Google data cannot observe directly. The series is still useful for this midterm design because it is standardized across cases, available at comparable weekly intervals, and interpretable alongside event timelines, related queries, trading volume, and policy events. It therefore provides a consistent signal of broader attention around each episode while leaving the limitation plainly visible: the paper is tracing external search salience, not reconstructing full domestic platform exposure.

For the Nongfu Spring case, the stock series covers January 2 to September 30, 2024, while the Google Trends series covers 40 weeks from December 31, 2023 to September 29, 2024. The search-interest variables are “Nongfu Spring” and “Wahaha.” For the salt panic case, Xuetian Salt and Suyan Jingshen stock data cover July 3 to October 31, 2023, while the Google Trends series covers 19 weeks from June 25 to October 29, 2023. The search-interest variables are “Japan nuclear wastewater” and “salt panic.” For the Urumqi case, the stock series covers reopening-sensitive firms — Air China (0753.HK), Haidilao (6862.HK), and Trip.com (TCOM) — over an October 2022 to February 2023 window. The weekly attention series centers on protest- and reopening-related search terms, especially “China protests” and “zero covid,” and is supplemented by related-query panels for “Urumqi fire” and “China protests.” The notebook-based case analysis uses 23 weekly observations and a dated policy timeline running from the November 24, 2022 fire to the December 7, 2022 release of the “New Ten Measures” and the later December 26 relaxation of quarantine rules. For the Lianhua Qingwen case, the stock series follows Yiling Pharmaceutical and Tongrentang from October 2022 to February 2023, while the weekly Google

Trends series runs from late August 2022 to late March 2023. The main attention variables are "Lianhua Qingwen," "Yiling Pharmaceutical," "China COVID medicine," and "COVID China drug," supplemented by related-query panels that help distinguish product-specific attention from broader reopening-era health-information demand.

C. Variables

The main independent variables are attention intensity and attention timing. In practice, these are measured through Google Trends levels, sharp week-to-week increases, related-query breakout signals, and narrative evidence that a topic became broadly visible on Chinese social media. The main dependent variables are market spillovers and policy spillovers. Market spillovers include price movement, abnormal trading volume, and evidence of investor-oriented search behavior. Policy spillovers include official announcements, import restrictions, supply-assurance statements, major policy reversals, and other institutional responses temporally aligned with peak attention windows.

The paper also treats case duration and event type as important contextual variables. Nongfu Spring represents a slower, reputational controversy, while the salt panic represents a short, high-intensity coordination shock. The Urumqi case adds a more explicitly political pathway in which the most consequential immediate spillover is not simply price movement, but central policy reversal followed by market repricing in firms expected to benefit from reopening. The Lianhua Qingwen case adds a fourth pathway in which reopening and health-information demand are associated with product- and company-specific pharmaceutical repricing, followed by post-burst normalization.

D. Data Processing and Alignment

Because the stock series are daily while Google Trends data are weekly, daily stock data were resampled to weekly frequency using week-ending Sunday averages for price, summed volume, and aggregated weekly returns. The weekly stock series were then matched to the Google Trends timestamps using nearest-date alignment within a seven-day tolerance window. This procedure produced 40 matched weekly observations for the Nongfu Spring case, 17 matched weekly observations for each salt-stock series, 23 weekly observations for the Urumqi case aligned to each of the three reopening-sensitive stocks, and 22 weekly observations for the Lianhua Qingwen case aligned separately to Yiling Pharmaceutical and Tongrentang.

V. Methodology

A. Analytical Strategy

The empirical strategy is a structured, case-based comparison rather than a large-N statistical test. The goal is to evaluate whether the observed sequences in four illustrative cases are consistent with the paper's threshold–pathway–spillover framework. For each case, the analysis proceeds in four steps: *(1)* identify the triggering event and build a timeline of attention-relevant developments; *(2)* locate the threshold-crossing window using the operational rule defined above; *(3)* compare the timing of threshold crossing with market indicators and policy responses; and *(4)* compare the cases to see whether distinct types of spillover are associated with distinct attention pathways. In other words, each case is read through the same sequence of questions about trigger, threshold, spillover, timing, and mechanism.

Within that structure, three analytical tools play different roles. Descriptive analysis summarizes event timing, baseline conditions, and burst-period changes in search attention, prices, and volume. Lag and correlation analysis are used more cautiously, mainly to check whether attention and stock series move together contemporaneously or with short leads and lags in the notebook outputs. Event-window analysis then divides each case into pre-crisis, crisis, and post-crisis periods so that differences in persistence, correction, and sequencing can be compared across cases. This design is appropriate for a midterm progress paper because it prioritizes transparency over statistical sophistication. It cannot establish causality, but it can show whether the timing and pattern of observed events are consistent with the conceptual expectations laid out earlier.

B. Operationalization of Threshold and Spillover

The threshold rule is used primarily to identify *when* a case enters a high-visibility phase, not to classify all possible online topics into firestorms and non-firestorms. The same language is applied across the cases: threshold crossing is placed where the two-of-three rule identifies a sharp attention break, contemporaneous visibility evidence, and an observable offline signal in the same narrow window. In the Nongfu Spring case, this point falls in early March 2024, when search interest jumped sharply above the recent baseline, the controversy was reported as viral on Weibo, and investor-related search behavior increased. In the salt panic case, it falls in late August 2023, when search interest jumped from near zero to the maximum normalized level, multiple related queries were marked as breakout, and market and policy reactions followed almost immediately. In the Urumqi case, it falls in the late-November 2022 window, when protest-related search attention surged, "China protests" reached the maximum normalized weekly value in the week of November 27, and the event sequence moved quickly from online pressure to nationwide protest visibility and central policy change. In the Lianhua Qingwen case, it falls in the two weeks after the December 7 reopening turn, when Google Trends surged, the timing was closely tied to reopening policy, and the strongest offline response appeared in the product-linked stock.

Spillovers are operationalized in two ways. Market spillover is identified through abnormal changes in price level, trading volume, and the timing of investor-oriented search activity. Policy spillover is identified through dated official actions or public institutional responses that occur during or immediately after the threshold-crossing window. The Urumqi case requires one additional distinction: in that case the main sequence is not simply trend to market, but trend to policy reversal to market recovery. The Lianhua Qingwen case adds a different distinc-

tion: the key sequence is reopening signal to medicine-demand attention to product-specific repricing and then correction. The relative timing of these outcomes is then used to assess H3.

C. Event Windows and Comparison Logic

For each case, the timeline is divided into pre-crisis, crisis, and post-crisis periods. The pre-crisis window establishes a short baseline for price, volume, and search attention. The crisis window begins when the threshold rule is satisfied and continues until attention and market activity begin to normalize. The post-crisis window captures whether effects dissipate quickly or remain persistent. In the Urumqi case, the crisis window is further divided into a protest-pressure period (November 24 to December 7, 2022) and a reopening-rally period after the policy reversal, because the case is analytically about a two-step spillover chain rather than a single shock. In the Lianhua Qingwen case, the event window is divided into a pre-reopening period, a medicine-demand burst from December 7 to December 25, 2022, and a post-burst normalization period beginning December 26.

This event-window logic serves two purposes. First, it helps distinguish abrupt attention shocks from longer-burn reputational crises and from cases that include both rally and correction. Second, it allows the paper to compare whether market responses appear contemporaneously with public attention while policy responses lag behind, which is the core expectation behind H3. The Urumqi case is especially useful here because it may represent a boundary condition: policy response was unusually fast, and markets appear to have started repricing before the formal announcement itself. The Lianhua Qingwen case is useful for a different reason: it shows that threshold crossing can be associated with short-horizon product-specific gains followed by correction rather than with only persistent damage or immediate normalization. Given the small number of cases, the comparison remains descriptive and interpretive rather than inferential.

D. Role of Survey Evidence

Survey evidence is not yet part of the current empirical findings and should not be treated as if it were already fully integrated into the results. In this midterm version, the survey instrument is included only as a planned triangulation tool and is reproduced in the appendix for transparency. The intended purpose of the survey is to capture awareness, perceived credibility, and self-reported behavioral influence for selected events, especially where direct platform-level exposure data are difficult to obtain. Until response data are incorporated and analyzed, however, the core findings of the paper rest on literature review, case timelines, attention proxies, and market/policy evidence alone.

E. Operational Challenges

Several challenges shape the analysis. The first is proxy validity: Google Trends captures external search interest, not native Weibo or Douyin visibility. This matters across all four cases, but especially for the Urumqi and reopening-era medicine-demand cases, where domestic platform intensity and recommendation dynamics were likely much richer than the public global search series can show. The second is comparability: Chinese and U.S. platforms differ in moderation, recommendation, and discovery systems, which makes direct cross-platform comparison difficult. The third is causal identification: attention spikes, market movements, and policy responses may be jointly driven by broader news shocks, macroeconomic expectations, epidemiological information, or strategic behavior by firms and regulators. For these reasons, the paper interprets temporal alignment as evidence of patterned association and plausible mechanism, not as proof of direct causal effect.

VI. Results

Each case subsection below follows the same sequence: event trigger and background, threshold evidence, spillover evidence, interpretation, and a brief caution about inference. This common structure is meant to keep the comparison focused on the same underlying questions rather than to treat the four episodes as unrelated narratives.

A. Case 1: Nongfu Spring Public Opinion Crisis

The trigger in Case 1 is reputational rather than regulatory or public-health related. After the death of Wahaha founder Zong Qinghou on February 25, 2024, online sympathy toward Wahaha quickly evolved into a boycott campaign against Nongfu Spring. By March 3, the controversy was being reported as viral on Weibo, and the event window then broadened through later corporate responses and financial disclosures.

Applying the same threshold rule used throughout the paper, threshold crossing is placed in early March 2024. Search interest for "Nongfu Spring" rose from a baseline of about 10 to 68 in the week of March 10, a 5.8-fold increase over the recent background level. Related queries also show rising investor attention: the query "nongfu spring stock price" increased by 190%. Together with contemporaneous reporting that the boycott had become highly visible on Weibo, these signals fit the pattern of threshold crossing rather than routine background discussion. Figure 1 links the attention breakout to the beginning of the stock-market decline.

The main spillover in Case 1 is a persistent market deterioration rather than a fast panic spike. As Figure 2 shows, Nongfu Spring's stock price ranged from a peak of HKD 47.95 to a trough of HKD 24.35 during the observation window, implying a maximum decline of 49.2%. Average price levels also deteriorated over time: the pre-crisis average was HKD 42.25, the crisis-period average was HKD 43.03, and the post-crisis average fell to HKD 29.95. Trading activity increased as well, with average volume rising from 4.2 million shares before the crisis to 4.6 million during the crisis, a roughly 10% increase.

The timing and persistence of the response suggest a slow-burn reputational mechanism: visibility rose first, the spillover appeared quickly, and the repricing then lasted far longer than the initial attention burst. In comparative terms, this is the

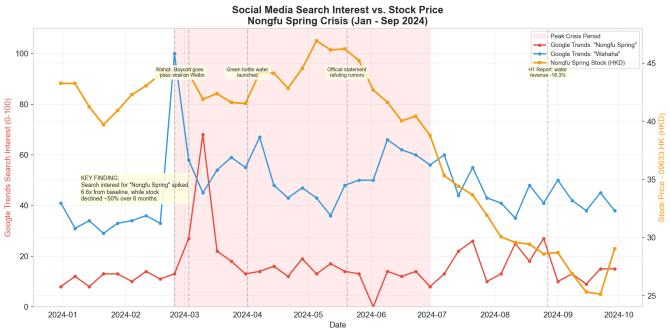

Fig. 1. Case 1: weekly search attention and Nongfu Spring stock price. The figure highlights the early-March visibility break and helps distinguish threshold crossing from the longer-run persistence that follows.

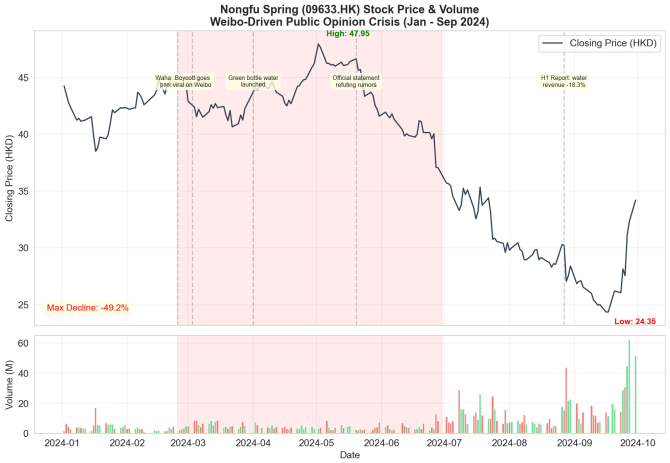

Fig. 2. Case 1: Nongfu Spring price and volume. The analytical value of the figure is that the downturn persists well beyond the trigger window, separating slow-burn reputational damage from a short correction cycle.

case with the weakest instantaneous spike but the strongest persistence. Even here, however, the paper treats the pattern as temporal association rather than proof that online attention alone caused the decline.

B. Case 2: Japan Nuclear Wastewater Salt Panic

The trigger in Case 2 is a sudden public-health scare tied to Japan's August 24, 2023 release of treated nuclear wastewater. Panic spread quickly across Chinese social media, triggering consumer hoarding of salt and sharp short-term stock movements in major salt producers. Government response also appeared immediately, including China Customs Announcement No. 103 banning Japanese aquatic imports, a China Salt Industry Group statement assuring adequate salt supply, and provincial warnings against hoarding and price gouging.

Applying the same threshold rule, threshold crossing is placed in late August 2023. Google Trends data show that search interest for "Japan nuclear wastewater" rose from 0 to 100 in a single week, the maximum value on the platform's

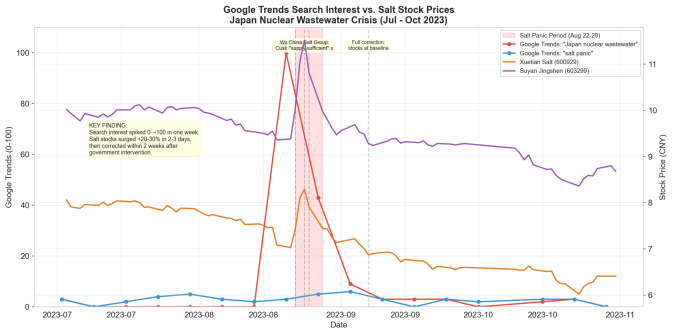

Fig. 3. Case 2: weekly salt-panic attention and salt-sector stocks. The figure's analytical role is to show how sharply the attention burst rises and falls, which supports the panic-coordination interpretation.

scale, then fell to 43 the following week and returned close to zero by October. Related-query data show that 12 out of 22 rising queries were marked as "Breakout," indicating increases above 5000%. The episode also coincided with immediate abnormal market activity and formal institutional response, satisfying the second and third threshold indicators in the same narrow window. Figure 3 shows the strongest and most abrupt threshold jump in the paper.

The spillover in Case 2 is both market and policy oriented, and both arrive quickly. As Figure 4 shows, Xuetian Salt's stock price ranged from CNY 6.02 to CNY 8.29, with a maximum decline of 27.4% from peak to trough across the study window. The pre-crisis average price was CNY 7.78, the crisis-period average was CNY 7.83, and the post-crisis average fell to CNY 6.66. Average trading volume rose from 7.7 million shares pre-crisis to 72.0 million during the crisis, an increase of 835%. For Suyan Jingshen, the stock ranged from CNY 8.36 to CNY 11.52, also implying a maximum decline of 27.4%. Its pre-crisis average price was CNY 9.85, crisis-period average price was CNY 10.68, and post-crisis average was CNY 9.10. Volume increased from 5.2 million shares before the crisis to 71.5 million during the crisis, a 1282% increase.

The mechanism suggested here is acute panic coordination: the trigger is sudden, threshold intensity is maximal, the spillover is immediate, and correction follows quickly. Unlike Case 1, both public sentiment and stock prices returned close to baseline within roughly two weeks, making duration itself part of the analytic contrast. At the same time, the paper does not claim that online attention alone generated the market and policy response, since the wastewater release itself was also a major exogenous news shock.

C. Case 3: Urumqi Residential Fire and Zero COVID Policy Reversal

The trigger in Case 3 is a politically charged crisis rather than a consumer controversy. After the Urumqi residential fire on November 24, 2022, online anger and mourning fed into broader criticism of Zero COVID controls. By November 27, white-paper protests had gone viral, and by December

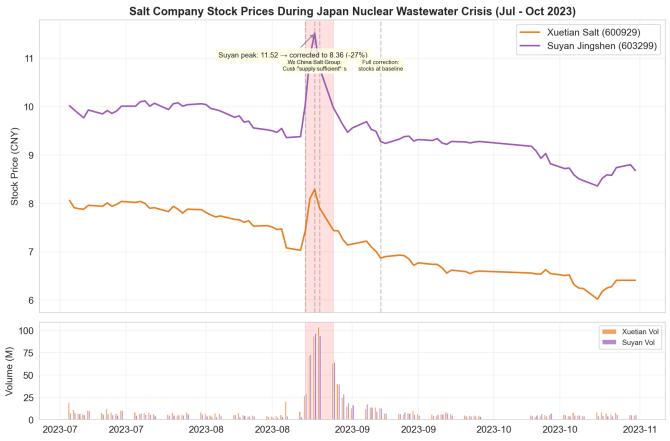

Fig. 4. Case 2: salt-stock price and volume. Extraordinary burst-period volume and the quick reversal help distinguish short panic speculation from the longer persistence seen in Case 1.

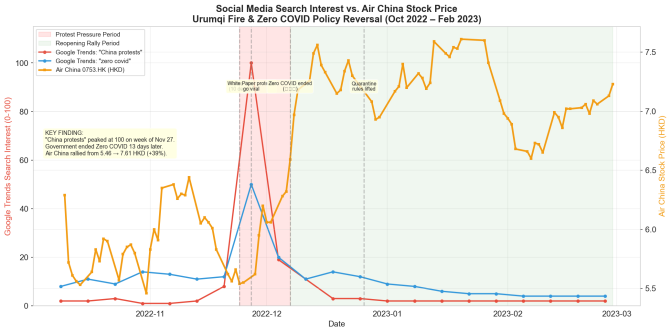

Fig. 5. Case 3: protest-related search attention and Air China stock price. The figure compresses visibility, policy reversal, and market repricing into one short window, which is why the case is read as a policy-pressure pathway.

7 the central government released the "New Ten Measures," widely understood as the effective end of Zero COVID. In the notebook analysis for this case, the relevant stocks are reopening-sensitive firms — Air China, Haidilao, and Trip.com — observed over an October 2022 to February 2023 window.

Applying the same threshold rule, threshold crossing is placed in the week of November 27, 2022. Protest-related search attention reached the maximum normalized level of 100 for "China protests," and the related-query panels for "Urumqi fire" and "China protests" indicate breakout-style escalation rather than routine background interest. Unlike the first two cases, threshold evidence here is not product specific; it comes from the convergence of fire-related attention, national protest visibility, and rapid movement of the issue onto the policy agenda. Figure 5 places the protest-attention peak alongside the Air China series and the December 7 policy reversal.

The spillover sequence is correspondingly different from the earlier cases. The notebook frames it as a three-stage chain: social-media pressure, then government policy reversal, then market recovery in reopening-sensitive stocks. The timing is unusually compressed. According to the notebook's summary, the December 7 Zero COVID reversal occurred just 13 days after protest search interest peaked in the week of November 27. Market indicators appear to have moved during the protest window and after the policy announcement rather than only afterward. Air China and Haidilao are described as beginning to rise before December 7, suggesting that markets were already pricing in the probability of reopening. On the announcement day itself, Air China traded 128 million shares, the highest single-day volume in the notebook's sample. Figure 6 shows that the repricing extended beyond Air China to other reopening-sensitive firms.

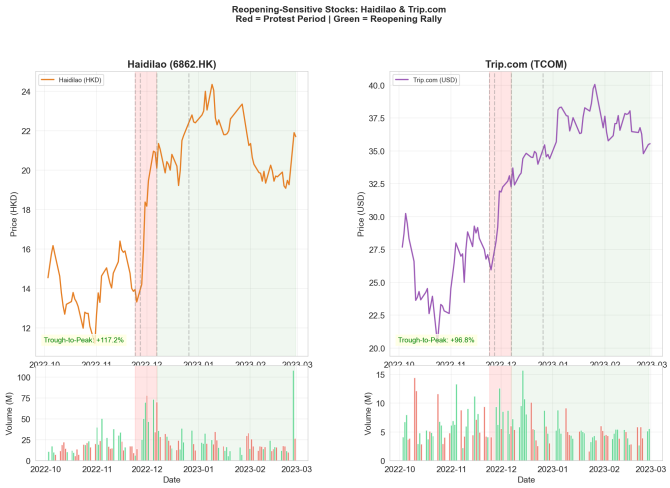

Fig. 6. Case 3: Haidilao and Trip.com during the protest and reopening windows. The figure's purpose is to show that the repricing extends across reopening-sensitive firms rather than staying confined to one stock.

The interpretation in Case 3 is therefore policy pressure rather than boycott, panic, or product demand. The most consequential spillover is central policy reversal, while the market response appears partly anticipatory and partly post-announcement. This makes Case 3 the clearest example of market and policy spillover operating through different but tightly connected channels. The notebook itself notes, however, that Google Trends captures global rather than domestic searches and that timing tests in a 23-week sample should not be treated as proof of causal effect.

D. Case 4: Lianhua Qingwen, COVID Medicine Demand, and Pharmaceutical Stocks

The trigger in Case 4 is reopening itself and the health-information demand it released. The case is therefore analytically distinct from the earlier three: it is not a boycott pathway, not a salt-panic commodity pathway, and not a protest-to-policy-reversal pathway. The relevant sequence begins with the November 11, 2022 "20 measures" that signaled easing, intensifies with the December 7 "New Ten Measures," peaks in the week of December 11 for "Lianhua Qingwen" searches, peaks in the week of December 18 for broader "China COVID

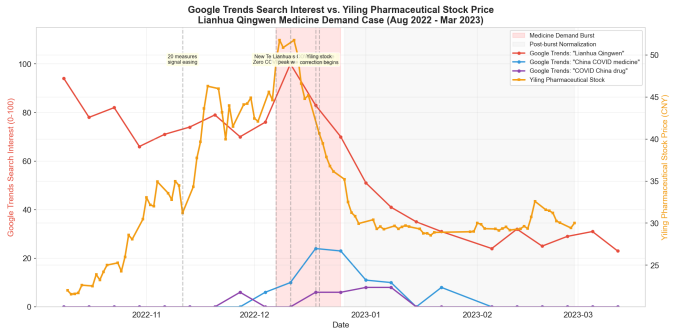

Fig. 7. Case 4: Lianhua Qingwen and broader COVID-medicine search attention alongside Yiling Pharmaceutical. The figure shows that the product-name spike sits within a broader reopening-era medicine information burst, which helps identify the mechanism.

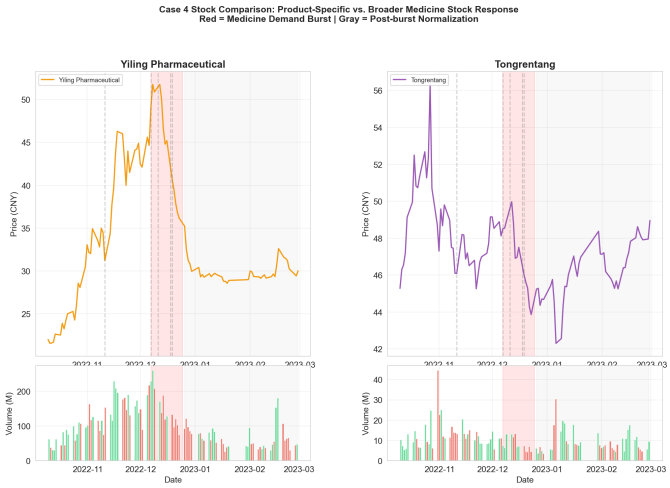

Fig. 8. Case 4: Yiling Pharmaceutical versus Tongrentang. The comparison shows stronger direct exposure and a clearer correction in Yiling, which is why Tongrentang functions as a useful comparator rather than a co-moving twin.

medicine" searches, and shifts into correction beginning on December 19 for Yiling Pharmaceutical.

Applying the same threshold rule, the case enters a high-visibility phase in the two weeks after the December 7 reopening turn. Threshold crossing is identified through the sharp Google Trends surge, its close event-linked timing, and the visible offline market response. "Lianhua Qingwen" reaches the maximum normalized search value of 100 in the week of December 11, while the broader "China COVID medicine" series peaks at 24 in the week of December 18. Figure 7 suggests that search attention is product specific but embedded in a wider COVID-treatment information environment rather than in a purely single-brand story.

The spillover evidence is more differentiated than in the earlier cases. Yiling Pharmaceutical is the main product-linked stock, while Tongrentang serves as a broader traditional-medicine and pharmaceutical comparator. As Figure 8 shows, Yiling's pre-reopening average price was CNY 33.68, its burst-period average rose to CNY 44.70, and its post-period average fell to CNY 30.00. Over the full observation window, the stock reached a peak of CNY 51.76 and a trough of CNY 21.56, implying a peak-to-trough range of -58.3%. Average volume increased from 118.5 million shares before the burst to 151.4 million during it, an increase of about 28%. Tongrentang moved much less: its pre-period average was CNY 48.59, burst-period average CNY 47.01, and post-period average CNY 46.22, while volume declined from 12.8 million to 8.2 million shares, a drop of about 36%.

The interpretation in Case 4 is reopening-driven medicine demand with product-linked repricing. It is not well described as a general pharmaceutical rally, because the strongest response is concentrated in the stock most directly tied to the product-name search shock. At the same time, it is not simply a firm-specific story, because the product-specific attention is nested inside broader reopening and COVID-treatment searches. What most distinguishes the case is that the spillover includes both a rally and a correction. The evidence is therefore plausibly linked to reopening-era medicine demand, but it does not prove that online attention alone caused the repricing.

E. Cross-Case Comparison

The four cases are analytically comparable because they vary on the same dimensions. Their triggers differ sharply: Case 1 begins with a brand-centered reputational dispute, Case 2 with a sudden public-health scare, Case 3 with a tragic event that becomes a national protest signal, and Case 4 with policy relaxation that generates a surge in health-information demand. Threshold intensity also varies. Cases 2 and 3 show the most abrupt normalized jumps, while Case 1 crosses threshold through a smaller but still clear breakout that remains salient, and Case 4 combines a maximal product-name spike with a broader but lower medicine-information series. This matters because the strongest immediate search surge does not automatically produce the longest or deepest spillover.

The main spillover form also changes across cases. Case 1 is dominated by persistent market damage, with little role for immediate formal policy response. Case 2 combines market speculation, consumer hoarding, and fast state reassurance in the same brief window. Case 3 is centered on policy reversal, with market gains appearing as reopening-sensitive firms reprice around that shift. Case 4 is centered on demand expectations, where the most directly exposed stock rallies more sharply than the comparator and then corrects. Duration helps separate these mechanisms further: Case 1 persists, Case 2 corrects quickly, Case 3 stays tied to reopening repricing, and Case 4 contains both a burst and a post-burst correction.

The timing comparison is equally important for the paper's argument. In Case 1, market spillover follows threshold crossing and persists well beyond it. In Case 2, market and policy responses arrive almost simultaneously. In Case 3, policy response is unusually fast and some market repricing appears

to begin even before the formal December 7 announcement. In Case 4, policy easing comes first, the search burst follows, and the stock response then moves into correction. Taken together, these differences suggest that spillover is not a single outcome but a family of responses that vary by trigger type, threshold intensity, duration, and the sequencing of market and policy effects.

Taken together, the four cases suggest a common pattern with different mechanisms underneath it. Episodes that satisfy the paper's threshold rule are more likely to be followed by observable offline responses than periods of ordinary background discussion, but the form of those responses varies substantially across cases. The results therefore fit a threshold-based salience argument more closely than a one-mechanism account of online influence. They remain descriptive and cautious, however, and do not by themselves establish direct causal effects.

VII. Discussion

The most defensible claim of this midterm paper is narrower than a causal theory of online influence. The four cases suggest that in China's governed attention environment, once an issue becomes visible enough to satisfy a transparent threshold rule, meaningful offline spillovers are more likely to appear than during ordinary background discussion. What varies is not whether spillover exists in exactly the same way each time, but which pathway it follows, how quickly it appears, and whether it persists or corrects.

This is what the four-case design adds beyond the existing literature used here. Prior work helps establish that firestorm thresholds differ across platforms, that information control in China shapes visibility, and that online attention can be associated with market or governance outcomes. The present paper contributes by putting those strands into one mechanism-focused comparison. Rather than treating "online attention" as a single variable with a single effect, it compares a reputational boycott pathway, a panic-coordination pathway, a policy-pressure pathway, and a reopening-driven medicine-demand pathway within the same threshold–pathway–spillover framework. That comparative structure makes the China-centered argument more precise without turning the project into a large-N or high-causality design.

The broader comparative implication should still be stated carefully. The draft does not directly compare Chinese cases with matched U.S. cases, so H1 is assessed mainly against the comparative baseline in prior scholarship rather than through original cross-national evidence here. What the cases can show is that a China-centered threshold framework is plausible and analytically useful. Case 3 is especially important in this respect because it highlights the limits of attention control under extreme public pressure: moderation may shape visibility, but it does not guarantee that politically salient signals can be fully contained once they become legible at scale.

The paper also remains limited in what it can claim. The cases are illustrative rather than representative, Google Trends is an external attention proxy rather than a native-platform visibility measure, and the observed sequences may also reflect broader news, policy, or economic shocks. For that reason, the findings are best read as patterned associations that help clarify mechanisms, not as definitive proof that online attention independently caused the observed market or policy outcomes.

VIII. Limitations

- The empirical design is China-centered and case based, so the paper does not provide a direct matched China–U.S. test.
- The four cases are illustrative and likely *most-likely* cases, which limits how far the findings can be generalized.
- Platform data are incomplete and not equally accessible across settings; Google Trends is only a partial proxy for Chinese platform visibility, and this limitation is especially important in the Urumqi and reopening-era medicine-demand cases.
- Measures of "threshold" and "spillover" rely partly on heuristic operational rules rather than direct platform observables.
- The Urumqi and Lianhua Qingwen cases use only 23 and 22 matched weekly observations respectively, so lead-lag and Granger-style results should be interpreted with particular caution.
- Causality is difficult to establish because attention spikes, market movements, and policy responses may be jointly shaped by broader news shocks, economic conditions, epidemiological developments, and strategic behavior.

IX. Conclusion

This paper develops a China-centered framework for studying how online attention becomes consequential in a governed visibility environment. Its main contribution is not to prove causation, but to show that threshold, pathway, and spillover can be used together to compare four different kinds of China-based episodes without collapsing them into one undifferentiated story of "virality." Across the four cases, visible attention is associated with offline consequences, but the relevant mechanism differs: reputational damage, panic coordination, policy pressure, and reopening-driven medicine demand do not unfold on the same timeline or produce the same kind of correction.

That framing matters because it shifts the question from whether online controversy exists in China to when it becomes legible enough to matter and what kind of consequence follows once it does. For the final version, the project would still benefit from stronger native-platform evidence, more systematic event coding across cases, and either one direct comparative case or a cleaner China-only framing if the comparative claim remains limited to the literature baseline.

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Appendix A
Survey Instrument

The survey instrument is retained as a planned triangulation tool for the final version of the project. In the current midterm draft, it is not yet used as part of the empirical analysis, and its role is limited to documenting the intended design for collecting awareness, perceived credibility, and self-reported behavioral influence around selected events.

Appendix B
Additional Figures and Tables